ADDITIVES

INTRODUCTION

Additives:

Any Substance that is added generally in small concentration to resins in order to :

- Alter their properties
- Facilitate processing
- Change the physical, chemical or electrical properties of end products.
- Reduce the cost

WHY TO USE ADDITIVES IN PLASTICS ?

- Improve processing conditions
- Increase resin's stability to oxidation
- Obtain better impact resistance
- Increase or decrease hardness
- Control surface tension
- Facilitate extrusion moulding
- Control blocking
- Reduce cost
- Increase flame resistance.

TYPES OF ADDITIVES

- Fillers
- Antioxidants
- Heat stabilizers
- UV stabilizers
- Colorants
- Impact modifiers
- Flame/fire retardants
- Cross-linking agents
- Blowing agents
- Lubricants
- Nucleating agent
- Plasticizer
- Slip & Anti Block Agent
- Processing aids
- Antistatic

FILLER

- It is used to modify mechanical properties & to reduce the cost.
- EFFECT OF FILLER
  - The use of inert fillers can influence the polymer properties in the following ways :
  - Increase in density.
  - Increase in modulus of elasticity.
  - Lower shrinkage.
  - Increase in hardness.
  - Increase in HDT.
  - Reduction of raw material cost.
  - For e.g. Calcium Carbonate

CLASSIFICATION OF FILLER

- Mineral fillers : e.g. $CaCO_3$, kaolin, feldspar, talc etc.
- Metallic, conductive & magnetic fillers : iron, copper, zinc & lead zinc & lead
- Fire retarding fillers : Sb_2O_3
- Organic and special purpose fillers : carbon black, wood, flour, walnut flour.
- Flake reinforcement : mica Flake reinforcement : mica
- Continuous filament reinforcement : fibre glass, carbon graphite filaments etc.
- Short fibre reinforcement : asbestos

HEAT STABILIZERS

- To prevent degradation that occur during high temp processing/fabrication.
  - Prevent Degradation
  - Absorb & Neutralize HCl gas evolved
  - Prevent Oxidation reactions
  - Prevent Discoloration
  - Displace active substituents with stable substituents

HEAT STABILIZER

- Lead
  - Tri Basic Lead Sulphate (TBLS)
  - Dibasic Lead Sulphate (DBLS)
  - Basic Lead Carbonate
  - Dibasic Lead Phosphate
- Organo-tin
  - Dibutyltin maleates
  - Dibutyltin bis mercaptides
- Cadmium / Barium /Zinc
  - Cadmium/Barium laurates
  - Cadmium/Barium/Zinc Complexes

UV STABILIZERS

- UV stabilizers in plastics usually act by absorbing the UV radiation preferentially, and dissipating the energy as low-level heat.
- Eg,

  Benzophenones – for PE, Polyesters

  Benzotriazoles – for PS, Polyesters

  Carbon black

COLORANTS

- Produces varieties of colored polymers
- Also improves
  - Mechanical Strength
  - Specific Gravity
  - Clarity

- Types Of Colorants
  - Dyes (soluble in polymers)
  - Pigments (insoluble)

COLORANTS

- Dyes :
  - Impart brilliant transparent color to clear plastics
  - Inorganic & Organic
  - AZOS for brightness and clarity
  - Anthraquinone (AQ) –Good heat transparent weathearibility (Auto tube light)
- Pigments :
  - Diarylide pigment
  - Lead chromate

ANTISTATIC AGENT

- Prevent electrostatic charges – mostly seen in PE, PP, PS, Nylons, Polyesters,

  Urathenes, Cellulosics, Acrylics & Acrylonitriles
- Because of insulation properties electrical charge may get deposited on the

  surface of the plastics produced during processing.
- This may cause severe damage to the products & equipments.
- It may cause accumulation of dust.
- The accumulation of static charge can be minimized by the use of antistatic

  agents.
- Chemicals added to plastics to reduce built up of electrostatic charges on the

ANTISTATIC AGENT

- Eg,
  - Amines
  - Quaternary ammonium compounds,
  - Phosphates
  - Polyethylene glycol esters

FIRE RETARDANTS

- Polymers being organic compounds burn or decompose on exposure to fire,

  generates toxic fumes, smoke.
- Eg,
  - Inorganic : Zinc borate
  - Halogenated : Chlorinate paraffins
- Applicable to polymers to be used in mines, automobiles, air vehicles, railways,

  etc.

CROSS LINKING AGENT

- Thermosets cure either on application of heat or cross linking/curing/ hardening

  agents.

- In some cases agent triggers off the reaction & therefore linking the resin

  molecules or It enters into the parts between molecules and completes the chain.

- Classification :
  - free radical initiator : Isobutyl nitrile, Alkyl peroxide .
  - non free radical agents : Isocyanate with polyol
- Used : Manufacture of Thermosets like polyurethane, Urea Formaldehyde, etc.

BLOWING AGENT

- Polymeric foams preparation
  - Mechanical
  - Chemically
- Substances called blowing agents, which on heating decompose giving N & CO_2.
- Applications: mattresses, pillows, insulation, foam pipes for non-p conditions.

e.g. Para toluene sulphonyl hydrazide.

LUBRICANTS

- Hot processing of polymers need lubricants either internally or externally.
- **Internal lubricants** → decreasing normal cohesive forces between polymer molecules → good flow conditions by lowering viscosity.
- They function by reducing intra-molecular friction before and during the melt formation of the polymer by promoting flow and reducing melt viscosity of the polymeric mass.
- They also enhance polymer properties like heat stability, impact strength, colour & clarity.

LUBRICANTS

- **External lubricant** don't let melt to stick to hot surface of processing. By forming a thin layer at melt-solid interface.
- Prevent friction between polymer melt & metal surface
- Prevent sticking to the metal parts
- They are usually high molecular compounds and have non-polar groups.
- Eg, Metal Stearates, Zinc Stearates, Calcium Stearates, Magnesium Stearates,

  Hydrocarbon waxes

IMPACT MODIFIERS

- To enhance the impact properties of certain brittle polymers so as to use in the field.
- E.g.
  - Chlorinated polyethylene EVA are used for PVC.
  - SBR is used for polystyrene material.
  - EPDM is used for PP (Bumpers)
  - Acrylic rubbers for Poyamides

ANTIOXIDANTS

- They inhibit or retard the oxidative degradation of materials at normal or elevated temperature during processing, storage or service due to
  - Heat
  - Light
  - Chemical Induction
- How it works? – It interrupt the chain reaction by combining with the free radicals forming a non reactive products
- E.g, Hindered Phenols or Aromatic Amines

PLASTICIZERS

- Improve process ability by reducing Tg
- These are high boiling non-volatile solvents.
- Polar with a high Mol. Wt. ester type organic compounds.
- Reduce internal friction between polymer chain.
- Effect of plasticizer
  - Easy melt
  - Improve flexibility
  - Increase Softness and Flexibility.
  - Improve Process ability.

PLASTICIZERS

- **Primary Plasticizers** – These are highly compatible with PVC and can be used alone.
- Eg, Phthalates, Di-Octyl Phthalate (DOP), Di Iso Octyl Phthalate (DIOP)

  Phosphates, Tricresyl Phosphate (TCP).
- **Secondary Plasticizers** – These are less compatible with resin & and are usually

  employed together with primary plasticizers.
- Eg, Di Octyl Sebacate (DOS), Adipic Acid Polyesters, Epoxidised oil.

SLIP & ANTI BLOCK AGENT

- **Slip** : Reduces coefficient of friction- They are high molecular weight fatty alcohols.
- Amount (0.05 – 0.2%) e.g. For film of 25 micron 0.01% of Oleamide.
- **Anti block Agents** : Prevent adhesion between the film surface Amount (0.05 -0.2%).
- E.g. – Calcium Carbonate in PVC film – Metal Salts – Fatty Acids – Natural & Synthetic Waxy Materials.

NUCLEATING AGENT

- Aiding transmission of white light in Plastics
- Forms large nos. of nuclei & reduces the size of spherulites.
- Eg. – Sodium, Potassium, Lithium benzoates. – Inorganic Powders – Clays, Silica Flour.

Thank You